

Annex 119A

(informative)

200GBASE-R and 400GBASE-R PCS FEC codeword examples

This annex provides example PCS FEC codewords produced by the 64B/66B to 256B/257B transcoding, scrambling, pre-FEC distribution, and Reed-Solomon encoding defined in Clause 119. This annex presents data in a tabular form. The contents of the tables are hexadecimal representations of the data and are transmitted to the symbol distribution function from left to right within each row starting from the top row and ending at the bottom row. The most significant bit of each hex symbol is transmitted first.

A constant transmission of Idle control characters is sent to the 256B/257B transcoder resulting in a continuous stream of 257-bit blocks, $tx_xcoded<0:256>$, shown in hex:

$tx_xcoded<0:256> = 007000000000000000780000000000000078000000000000007800000000000000$

In this example, an alignment marker is due for insertion and the scrambler seed [see Equation (49–1)] just before the 257-bit block was scrambled was:

$S<0:57> = 24E6959D0FA5DBD$

After scrambling and inserting the alignment marker group am_mapped , the variable $tx_scrambled_am$ is produced as shown in Table 119A–1 for 200GBASE-R and Table 119A–2 for 400GBASE-R. In these tables, italicized characters are alignment markers and bold characters are padding for the alignment markers. Just prior to insertion, the seed of the PRBS9 generator used for padding was $P<0:8> = 0x100$.

After distributing into two message blocks, and PCS FEC encoding, codewords c_A and c_B are produced. c_A and c_B may be illustrated as two 5440-bit representations cx_A and cx_B as follows:

For all $i=0$ to 543:

$cx_A<(10i+9):(10i)> = c_A<i>$

$cx_B<(10i+9):(10i)> = c_B<i>$

The expanded codewords are shown in Table 119A–3 and Table 119A–4 for 200GBASE-R and Table 119A–5 and Table 119A–6 for 400GBASE-R (bold characters are parity).

The example codewords have $tx_am_sf<2:0> = \{0,0,0\}$.

Table 119A–1—Example $tx_scrambled$ with alignment marker group for 200GBASE-R

Index $<i:j>$	$tx_scrambled_am<i:j>$
$<0:256>$	<i>0B2ACAB2ACAB2ACAB2ACA9324C9324C9324C942108C22C90A2697A225314</i>
$<257:513>$	<i>1AA6A9BA6E98A629AB6ADAB6ADAB6ADAB6AD9BE6D9B26DDB76E5B46DEF365BDF1</i>
$<514:770>$	<i>0C42A493A18D4F681A02679E4456DD1C98CFC35417A7B1742DB5334A1BD2041B7</i>
$<771:1027>$	<i>07ACE48B2FD61333BFB75C3DC524204CEBD0955D1B04967A00846139561BD3720</i>
$<1028:1284>$	08CCD177010357E8DAB65B6F9160968075341AC958CFC828D8372DC66242646D8

**Table 119A–1—Example tx_scrambled with alignment marker group
for 200GBASE-R (continued)**

Index $\langle i:j \rangle$	tx_scrambled_am $\langle i:j \rangle$
<1285:1541>	026FEAA09736E915DD54C5F7E600ECFCBD7D7C207941C5D7182D8B95FE412ED12
<1542:1798>	014CCE446415F10CDC7BBAFB1CC4B442EF5737AFB4F3BED48D240C9095F5FD3A6
<1799:2055>	10219FBE135EDE0B73C15238C150007886562E30A50CB27DF341954DB8CA1C564
<2056:2312>	16598EAD3C229838CB9BEE7F793139EE1F89FDAD9071687CBEDF89B4E527A03CD
<2313:2569>	1C2044F260A85EE6EB506C25E7C1196298548B4B3576F1B084C83F2E3CB5FC5F6
<2570:2826>	1C9D40EF27471536FE9EB5E3BC3870D46755080EA6D29F09CF8FA697A738DD3EC
<2827:3083>	193FA8A7675CE03706A49019B93544E19DD66CE78E6A94BED07ECCA7DFD2076D
<3084:3340>	081919F234C00959E5AFFC9F83C909805378B5F3F2E4917F3E48B5DE4758CEA42
<3341:3597>	1D378C193CDD3D0F781ABC3529BF76A66B7C73A7230B517D96DAFF6A39F9EA9B6
<3598:3854>	05E852B6917E9D1410B6509E778629A516FB2BCEABA79F12241DBC94CDAFFFFF02
<3855:4111>	1D98F5991744C9FC5393EFD6299558CB4F28DF3BF3C8636DBA505E3A296C7BD2E
<4112:4368>	00FAACFBFDBC1C29CED046C73C9AAAFE3DE8849ADAD16E864814846BB8CB889AD
<4369:4625>	04D11BFE0E8DEF5ECFDBE45D1EE4607B4B44DF874F8E976DD828FCFD38158A8FF
<4626:4882>	00EE28B635049BA9506024BA13862A1D6BAE22D0DB5DDB1F4DBE0F80E9EDBBCC6
<4883:5139>	1CD0769BC76446DA04136E7C6D19905DFB81AC3BFD931426901028D729E529F5A
<5140:5396>	14C3FF33243D7D2B5137B63359FD25B82CDE769D0F10D2E60A0386E60820BE8F4
<5397:5653>	1993F1FE2D6C60459F26A74BD0265C2959899E4C5B24196AFD51DB241CFCF9091
<5654:5910>	18EE61EA5A0D584DE8AC602618874BBA60A907115513BC960F6BE32C68F1F2458
<5911:6167>	042251FD6EF16662F5C99D9779B20B865B1401C96034C99550401BCB2792E5638
<6168:6424>	0CE3AE06476B769F33E5577CE4BA6D62071CABE3EA9561D6077FD23909568A25A
<6425:6681>	1BBC46AC3DF26A5BB0EA4FDBCBF42738A84007619C594E24AB396AFB5F05FBFF
<6682:6938>	076C96A407E106D494AA6B0C5168E1F335B812D7BC530CE8C6FC13F6C54EC2165
<6939:7195>	056CC06AFE2BD2A9894C4D1AD9E632F861208AD3894EFF0DE8B0291C486EA3281
<7196:7452>	094534A897031E6307E32C1906C85E99CD4696FC818D2B5EAF26A576DE149AE12
<7453:7709>	085AE9048EDDC5401827FAA937411FF35C10287236FFDC87E7E7E334B038EE3F9
<7710:7966>	014179E0E2359D3B961A1302FA4B7AC384F0564B15D7504DADBE3C65EE8F6F13A
<7967:8223>	08D40F4B092621CCA4919E81D0C15014204622D41814894DBE389C149E5E2257A
<8224:8480>	0E4772C6718DA5361ABFAAD60F5C32D304740DE671045CDAD6617125F4BBF457D
<8481:8737>	071A7D2B02B41F13971A22FE8A29F0A03C1CEC43CA0411D7872F18D12E0A2BD06
<8738:8994>	1344403844B360B4A61968D045941FF821511C2E95455C300EED811D314AD7018
<8995:9251>	0275A5C4A36F6B8B13AFAFFCFDFBD9B13F00488494886CDEB13B31FC9F7941CF
<9252:9508>	168E51B4882712DEC042307B45EBFF549E07C92FD3C3DAB5A655CC419BBA06F1
<9509:9765>	02B88C8D61954691B36009D54637647F750E1B99733303EA76E0805B182D3B9D0
<9766:10022>	1C50996262A56FB5D7631247C2F52B5150C17BA61FEB56A37BE13F2ABC226AA0A
<10023:10279>	1A1F1CF493816E6F69603FF83E895BA5AFF2ECB8E9090CB28AE93C232756FEDAF

**Table 119A-2—Example tx_scrambled with alignment marker group
for 400GBASE-R**

Index < <i>i:j</i> >	tx_scrambled_am< <i>i:j</i> >
<0:256>	0B2ACAB2ACAB2ACAB2ACAB2ACAB2ACAB2ACA9324C9324C9324C9324
<257:513>	192649926499264991B421184592144D2F445014D517435064290242CA669AA6A
<514:770>	1374DD314C5354D5354DD314C5334DD356D5B56D5B56D5B56D5B56D5B56D
<771:1027>	0B6ADAB66F9B96C9B76DDB96D1B76D1BF6E1B26CDBD6D9B8B8196F7C588549274
<1028:1284>	063530AC615723489C7AA5333D91C8099E79115B747311F9C333AE74ADEAEFBF9
<1285:1541>	1D7E1AA0BD3D8BA16DAAB83C86F1C6A4C3D959E99A91DE9020DB9EB3922FA719F
<1542:1798>	036FA98F2DAB3957B8B0999DFDBAE1EE2924183BB157354862E1154102985E84A
<1799:2055>	15D1B04967A30D5178CE48F0569849A00846139561BD37228569FB24B7E4D4CC0
<2056:2312>	08CCD177010357E8DAB65B6F9160968075341AC958CFC828D8372DC66242646D8
<2313:2569>	026FEAA09736E915DD54C5F7E600ECFCBD7D7C207941C5D7182D8B95FE412ED12
<2570:2826>	014CCE446415F10CDC7BBAFB1CC4B442EF5737AFB4F3BED48D240C9095F5FD3A6
<2827:3083>	10219FBE135EDE0B73C15238C150007886562E30A50CB27DF341954DB8CA1C564
<3084:3340>	16598EAD3C229838CB9BEE7F793139EE1F89FDAD9071687CBEDF89B4E527A03CD
<3341:3597>	1C2044F260A85EE6EB506C25E7C1196298548B4B3576F1B084C83F2E3CB5FC5F6
<3598:3854>	1C9D40EF27471536FE9EB5E3BC3870D46755080EA6D29F09CF8FA697A738DD3EC
<3855:4111>	193FA8A7675CE03706A49019B93544E19DD66CE78E6A94BEBD07ECCAE7DF2076D
<4112:4368>	081919F234C00959E5AFFC9F83C909805378B5F3F2E4917F3E48B5DE4758CEA42
<4369:4625>	1D378C193CDD3D0F781ABC3529BF76A66B7C73A7230B517D96DAFF6A39F9EA9B6
<4626:4882>	05E852B6917E9D1410B6509E778629A516FB2BCEABA79F12241DBC94CDAFFFFF02
<4883:5139>	1D98F5991744C9FC5393EFD6299558CB4F28DF3BF3C8636DBA505E3A296C7BD2E
<5140:5396>	00FAACFBFDBC1C29CED046C73C9AAAFE3DE8849ADAD16E864814846BB8CB889AD
<5397:5653>	04D11BFE0E8DEF5ECFDBE45D1EE4607B4B44DF874F8E976DD828FCFD38158A8FF
<5654:5910>	00EE28B635049BA9506024BA13862A1D6BAE22D0DB5DDB1F4DBE0F80E9EDBBCC6
<5911:6167>	1CD0769BC76446DA04136E7C6D19905DFB81AC3BFD931426901028D729E529F5A
<6168:6424>	14C3FF33243D7D2B5137B63359FD25B82CDE769D0F10D2E60A0386E60820BE8F4
<6425:6681>	1993F1FE2D6C60459F26A74BD0265C2959899E4C5B24196AFD51DB241CFCF9091
<6682:6938>	18EE61EA5A0D584DE8AC602618874BBA60A907115513BC960F6BE32C68F1F2458
<6939:7195>	042251FD6EF16662F5C99D9779B20B865B1401C96034C99550401BCB2792E5638
<7196:7452>	0CE3AE06476B769F33E5577CE4BA6D62071CABE3EA9561D6077FD23909568A25A
<7453:7709>	1BBC46AC3DF26A5BB0EA4FDBCBCFB42738A84007619C594E24AB396AFB5F05FBFF

**Table 119A-2—Example tx_scrambled with alignment marker group
for 400GBASE-R (continued)**

Index $\langle i:j \rangle$	tx_scrambled_am $\langle i:j \rangle$
<7710:7966>	076C96A407E106D494AA6B0C5168E1F335B812D7BC530CE8C6FC13F6C54EC2165
<7967:8223>	056CC06AFE2BD2A9894C4D1AD9E632F861208AD3894EFF0DE8B0291C486EA3281
<8224:8480>	094534A897031E6307E32C1906C85E99CD4696FC818D2B5EAF26A576DE149AE12
<8481:8737>	085AE9048EDDC5401827FAA937411FF35C10287236FFDC87E7E7E334B038EE3F9
<8738:8994>	014179E0E2359D3B961A1302FA4B7AC384F0564B15D7504DADBE3C65EE8F6F13A
<8995:9251>	08D40F4B092621CCA4919E81D0C15014204622D41814894DBE389C149E5E2257A
<9252:9508>	0E4772C6718DA5361ABFAAD60F5C32D304740DE671045CDAD6617125F4BBF457D
<9509:9765>	071A7D2B02B41F13971A22FE8A29F0A03C1CEC43CA0411D7872F18D12E0A2BD06
<9766:10022>	1344403844B360B4A61968D045941FF821511C2E95455C300EED811D314AD7018
<10023:10279>	0275A5C4A36F6B8B13AFAFFFCDFDBD9B13F00488494886CDEB13B31FC9F7941CF

Table 119A-3—RS(544,514) codeword A for 200GBASE-R

Index $\langle i:j \rangle$	$cx_A \langle i:j \rangle$
<5439:5120>	A6A9AA6A9A64992649921486284BD26519565D946D5B56D5B57D936ED8B6CF7EFA842EDE6024F1DB
<5119:4800>	63156BC9A159A5EDB273FD0CCBB63B810BD2EA92401C8F6204458108BF4D44C2D1675209D8331CC8
<4799:4480>	40D556CEEAA30193F3AF9E2E8B43EA7494144FD4767AE467A1D9F9656C125225F1007DEDCE7100A
<4479:4160>	84474C2BBEA631DA8F34E5832CE9FCE47F0ADD1CE9F2325278204197782B7A662A1B4BF6C844FEB4
<4159:3840>	6F8155CB65AF1EE61D57AE3CA3E74B6399B457AE3B092274729AF1E4A60BD4C4F0ADF3018F39FE78
<3839:3520>	00DA3A7FD18904FB9B2E9832E1EF0FFD999CE031BEBD69C6552F4B672E84F2230D15EACBA247B6D9
<3519:3200>	1FF19A27932706B6A8F273E1376C7ADA74BAAF7FA1C16E72AABDEB2B4424B13A65AF62171EB7D9E2
<3199:2880>	E048B2E3A541AFC1A83FA290552920391C28EAD81B7EDE02EDCED9689A059D98BEEB586FE420214E
<2879:2560>	E5065667D74558D5FE83B711E338380822FC991F0613ED2E999A89E24FD5DBBCE90B3BB490D6A032
<2559:2240>	A5E8311EE4E0E631E0D129376199937B9D0A36934CA0A4F749870EBB71F25524E35A71BE30DF7138
<2239:1920>	8B76B5613EDDBC9FD3390226172B35DF7E8DCD2B10D491ACB499E40478B907D1B0876AD58BDA458B
<1919:1600>	19C61D4729EC1408494C515223C4FC4C34259FED58EAF5290F439D6710A8E4325F8A0E4E2FFE7F31
<1599:1280>	1C27F3D223B942CBE9AF1E46915EDA4C7784BAF092533E630BA006260A9171D2811CE51A763C36AB
<1279:960>	D79968151CD9D0C7D05F8E75282FA7F4651C04DC053AE8F5D2BD516438360C85805020871150180D
<959:640>	518076B85275B5CFFDFBF21124497B336F9CE14E08B69105BCFE8F2A4B872D467EED1E3126189401
<639:320>	62B8970BA6C0DDBD432D70B4C4A9D691BA115B41F0EB50FAAAAC85ABCE815BFFC45DA5DD09734C97
<319:0>	9337F1A9130B2CF27EA48EB65E77EADCEAB99C502FBC45E7F255A8ADA28CCA6D50F528D5740A36F4

Table 119A-4—RS(544,514) codeword B for 200GBASE-R

Index $\langle i:j \rangle$	$cx_B \langle i:j \rangle$
<5439:5120>	A6A9AA6A9A6499264992109A2C8A226519565D946D5B56D5B56DBB69DBB66986849163C0BCC51271
<5119:4800>	1FBD07455BC28405E74486BF7C3892732A40D979621AAEC9981DF56AB7B620D706B398629A748DB1
<4799:4480>	FDA9025EABFBF70AF443E098E7442722E6613211EF31B1695EFAFB958926D7ECBF9AB2D06A083380
<4479:4160>	6A5434D0B3B2A14D26B88858EEFA4FEF3C81370BF6F960BC3D3CA155D8360F94644AAC36013F1E3F
<4159:3840>	C9DEE472FE3D438314215B3212F9CBE5FF26E41C562602AEE673559D7CDFBE5B813162523AD3F084
<3839:3520>	D967D24FE7758D784B1ECF30B5614A5BBED394ADDA56FE7C6DA53D11454DF70B26FAAE7CF0494BFE
<3519:3200>	BA26B22E8FF7E65D32C5CFDB10A517DE381BE41EE71B1B263F421ADC2C943B8882C87FDEFF3E8A27
<3199:2880>	4B5FBC7FB6FC41CC54EE637B2304BAA33AD5A37597FE0BC867E0B8FB632063C261DF708C825B169E
<2879:2560>	5AFFE84AA5BD8D6B4BD9172580A33BA0997E3F6B68A5602D43645A3A60AB091F38E2F0582F603023
<2559:2240>	2EC1255527EB6C627D08BF9A3AF66E4DA9880301A9A00E4DA98E530DDBE67DF2EC0BEA5283597D48
<2239:1920>	48A3DF0E565C3DB0B551B81A3491693EBFD49BC05B0AA51B0EED7D6866C67E6E55340C47E552326D
<1919:1600>	F4C121CBFDD14E2574A0A5A0718DA66C0CB2C4095AB276DEC984A09EED8057C2EB3D40DB213F9CD2
<1599:1280>	1DC5083F3534C196903835175D938FBBD1E2B03484649816A0042D12836C3929EEA7711CE52FA835
<1279:960>	98710CF510D6A47BA5F5F954023C5845D0FE0FC2208E12C450580046925AB4F44FFD154B875BB971
<959:640>	5A9033A7DB8D3D7FB36600E48363233F8A72DD6C1C99B18BD155DF097EAD54CC1C09D4AC62AB36B9
<639:320>	BB2EF760CC95C100639D850C8FB4DDF1295854BB6BFD87E48782A3EC919D06B07DABFD1D0245150F
<319:0>	6AFD68CE7558FB5C8911150C62F97B29D912BCA53CCCF26F707B9AF277C6FD62055979796D2B8F85

Table 119A-5—RS(544,514) codeword A for 400GBASE-R

Index $\langle i:j \rangle$	$cx_A \langle i:j \rangle$
<5439:5120>	A6A9AA6A9A6A9A6A9A6A9A649926499264992D886284BD280BA2609026519565D946559565196
<5119:4800>	6D5B56D5B56D5B56D5B57D936ED8B62D876CD9B6077EFA842E86550894ABDE6024F1DBC4598972EA
<4799:4480>	BCD56BC9A11D7D8256A659A5EDB273E5F63E326A3BCCCB63B3054625443810BD2EA92C331E3C619
<4479:4160>	401C8F61447E7EDCCA66077D56AFED4815C9AFE6D8B691236C7F7A4894AA7ECDCEBC10380639D209
<4159:3840>	48B9904D84BBD645B57BEBEE96049F5CB27E7ACB40A8A0DE01AA509342C6CB8534AAEA22633BD93
<3839:3520>	3BFF204FC2FD8E502D0F4F285D750D83E51D13AB0DB043C68F7267B51CBF8F70ECC40896DC8CBE72
<3519:3200>	F95FC9B80795988ABB99CD56F5F37EFB6EC4E58D4BEB0FF21F679F490F9DD635A10C7B3FC2D5A52
<3199:2880>	56CFB4E62BB695B79D1BA97F4050533DD2C9BEAB1F1C1653FF6E99AC88A3FDE99B4FB1F3F6C82945
<2879:2560>	F7BE0EFB07B9C6CACB8F9096B308258EF228B01FF7AFCBA189D2C7E71EED3F207F163B58EEC0C02E
<2559:2240>	A8EEB169DD65CF86F319B82E36DAC8D8E09077DCA3009AC4A7569FFE11A96F735ED0F685E96028CE
<2239:1920>	E8265F8CDA9A39540A509936869A2A8267CE3ABC560BD80F08CBB0415749FACB149C42AFE68EBE9B
<1919:1600>	135A6E02C0AA5807906AE384CB74F99F6CB700FA5480D25D52D208FBC295570F642C546E2685255A
<1599:1280>	0FBFFD27F056F2A147C33B6F561BB1DF9B994E0351D95C8F9B7D104470FF3463815D28A958106069
<1279:960>	9B3324B302D6AC9DB6B2A10823B86055E0BACC50F6F847E434C7414E0DCD0D206DA50E4D75DB65E3
<959:640>	2EE470AD0DE1092E04A8812B44A3DB0E7A77A9DCC719FEA0DA61C433F4435891EE97DBE750C8D16

Table 119A–5—RS(544,514) codeword A for 400GBASE-R (continued)

Index $\langle i:j \rangle$	$cx_A \langle i:j \rangle$
$\langle 639:320 \rangle$	51543380F0881388B114962019A6962D0D13FC4552E1D6EE5CD6A408EAF6237F5FECD980292CD9C8
$\langle 319:0 \rangle$	4FC29E524AC56D33248BE7CBB1EBA5F71F5D872E35E9AFA1B54DB4D1A4FC09E8057F65FBC3E55FD5

Table 119A–6—RS(544,514) codeword B for 400GBASE-R

Index $\langle i:j \rangle$	$cx_B \langle i:j \rangle$
$\langle 5439:5120 \rangle$	A6A9AA6A9AA6A9AA6A9A649926499264992109A2C8A22ACAC250B426599565D946559765197
$\langle 5119:4800 \rangle$	6D5B56D5B56D5B56D5B59DBB69DBB6FD936BD8766986849163C6A27C7265713CC5127173FAEF6BEF
$\langle 4799:4480 \rangle$	1FBD07455B4F238F0D79E24405E744E62BEDA55386BF7C3892EE19DA30550CAA40D9798A89C6A164
$\langle 4479:4160 \rangle$	621AAEC9A824EC9203458118BB4F4402C1635209D83314CA401576CEEBA30193FFAE9EAE9B4FEB74
$\langle 4159:3840 \rangle$	940447D476FAE46BA3D97965681352E5E1087DED3CC7100984C74C2FBEA631DA8335E5832CEDFCE4
$\langle 3839:3520 \rangle$	3F1ADD1FE9B232567A2041A7782A7AA62A1348F6C874FEB76F8145CB65AF1ED61554AE7CA3EF4B63
$\langle 3519:3200 \rangle$	999453AC3B093278719A71F4AE09D404C0A9F0014F09F279005A3A7FD289C4EB9B2C98F2C1EB0EFD
$\langle 3199:2880 \rangle$	599CE431BEFD79C6542F4B572684F2A32D19EBCB2267BAD81FF1BA2F93274686A4F37361176C78DA
$\langle 2879:2560 \rangle$	749AAB7CA1415E7EA8BD6B3B4026B13A65AF6317DEA7DDE1E0C8B2EBA5412FD1A83DA29055292039
$\langle 2559:2240 \rangle$	5C08E6DB1B3EEE0EECED94892079D98BEE35A6F6410294FE5866667D64598E5F680B711C33C3A08
$\langle 2239:1920 \rangle$	E2FC9D1E0693FD2E999AC9C24FD4DBFCC90338B490C6AC33A5A8111EE5E0A631E0D229F75199927B
$\langle 1919:1600 \rangle$	1D0A36924C20B4F749870EAB79F355A4E35A72BE70FF713A8B76B5693CDDFC9FD73802E6072736DF
$\langle 1599:1280 \rangle$	7EADC1281094A1A8B599A41478B9075190836BD54BCA458B19862D4729EC9418414E51D233C8FC4C
$\langle 1279:960 \rangle$	F41597EE58EAF5250E435D7710ABE4B27F820C4EEFFE7B331C27F3D623B942CBE1AC1E46815EDB4C
$\langle 959:640 \rangle$	B784B2F192531E6308A04606029071929114E71AB62C36A9D71958111ED9D0D7D45D8E353823A7F4
$\langle 639:320 \rangle$	E53C08DE057AD8F1D0BD114438370C05B05422875160140E51C056BC5075F5CFFBDBF211044D7A33
$\langle 319:0 \rangle$	EFBCE079F39FDD0E6BDD2B8ED1CC0790B29C992D54C852DDD29825193C8CAEE6F675E83ECA4495AD